



BYD Microelectronics Co., Ltd.

BF20A6 芯片应用和设计指导

1/10 Inch VGA SOC CMOS Image Sensor With DVP and SPI

BF20A6 芯片应用和设计指导



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1 外围电路

1.1 DVP 接口

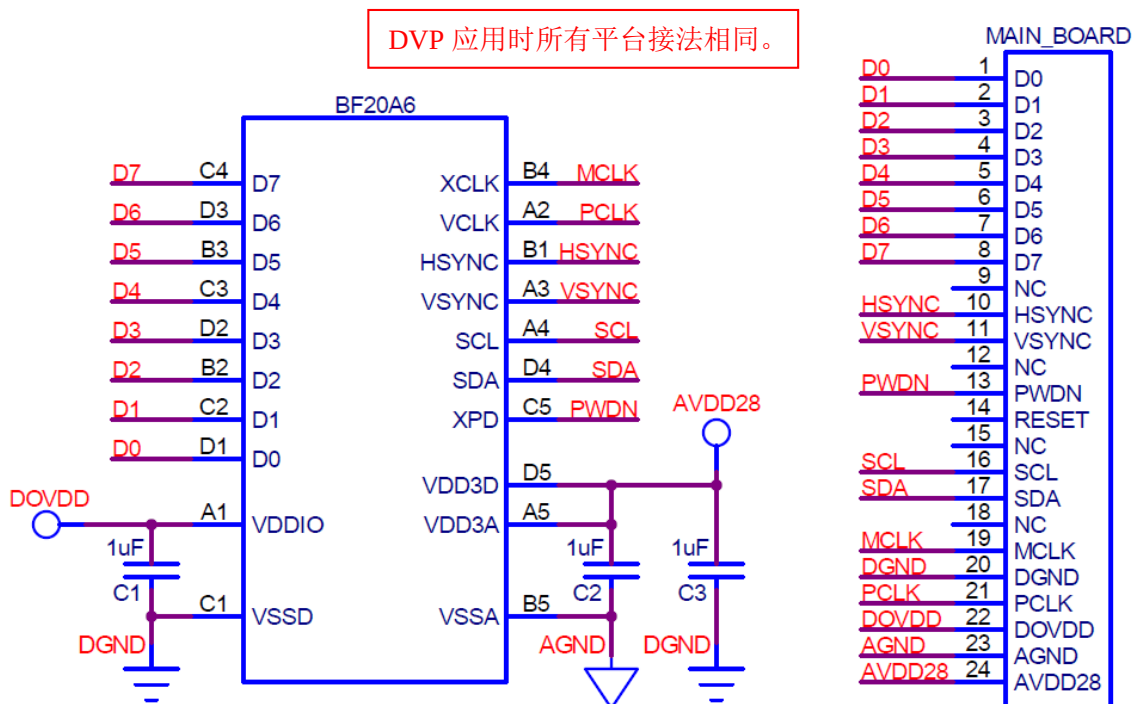


图 1-1 DVP 接口外围电路图

1.2 SPI 4-bit 模式

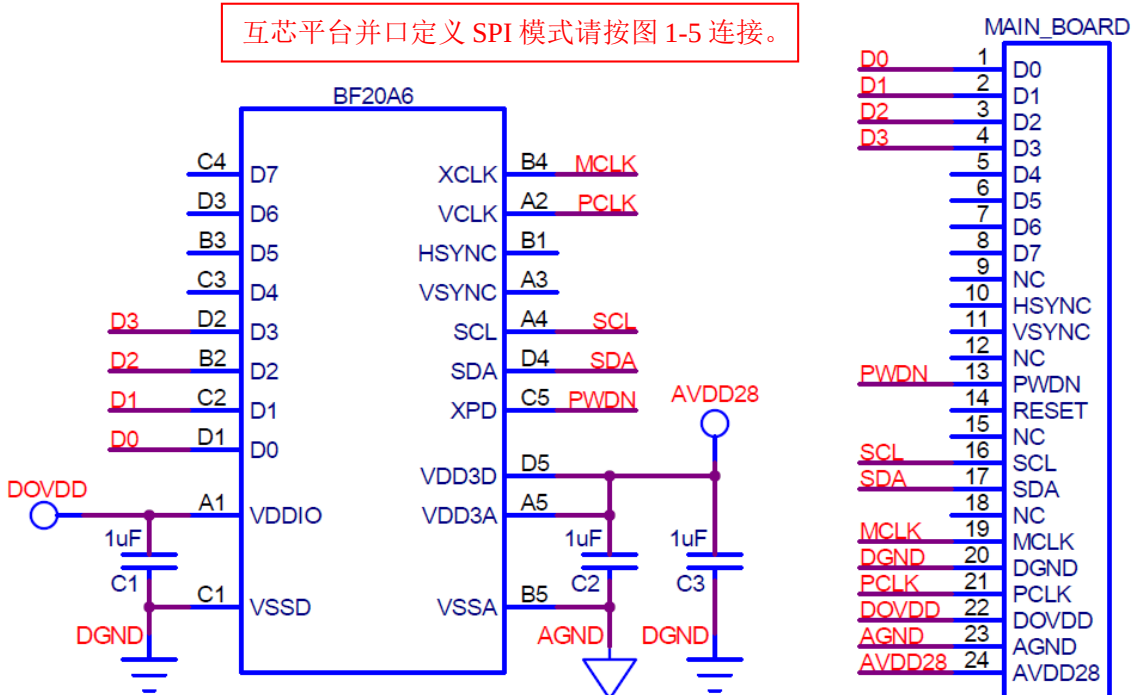


图 1-2 SPI 4-bit 模式外围电路图

1.3 SPI 2-bit 模式

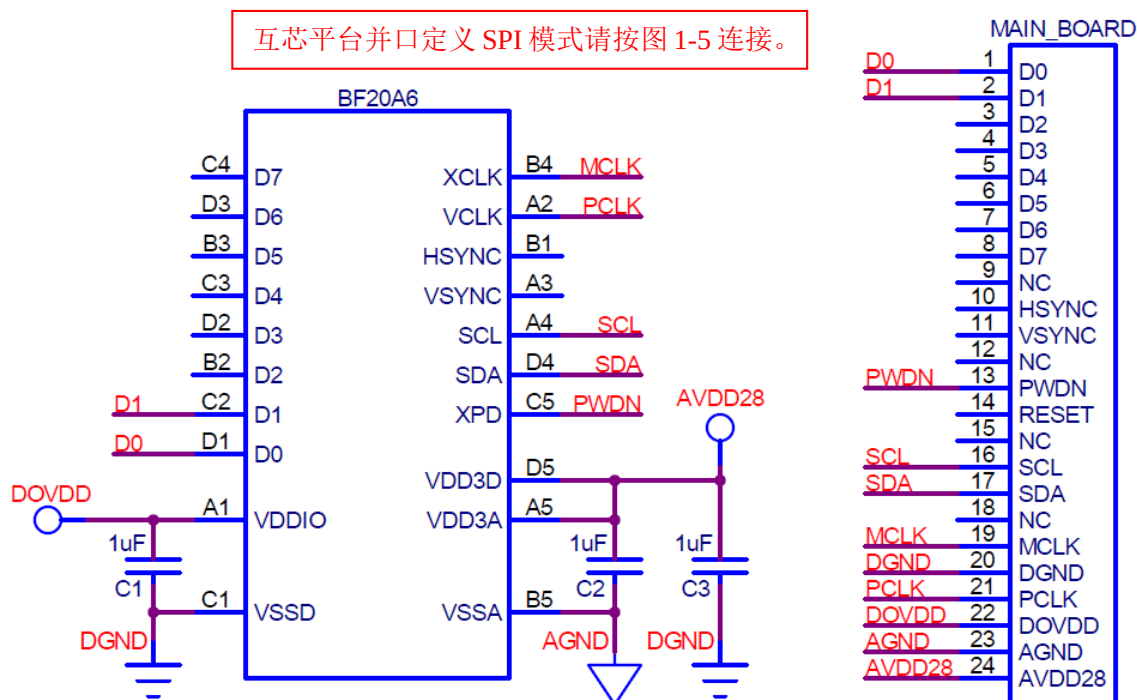


图 1-3 SPI 2-bit 模式外围电路图

1.4 SPI 1-bit 模式

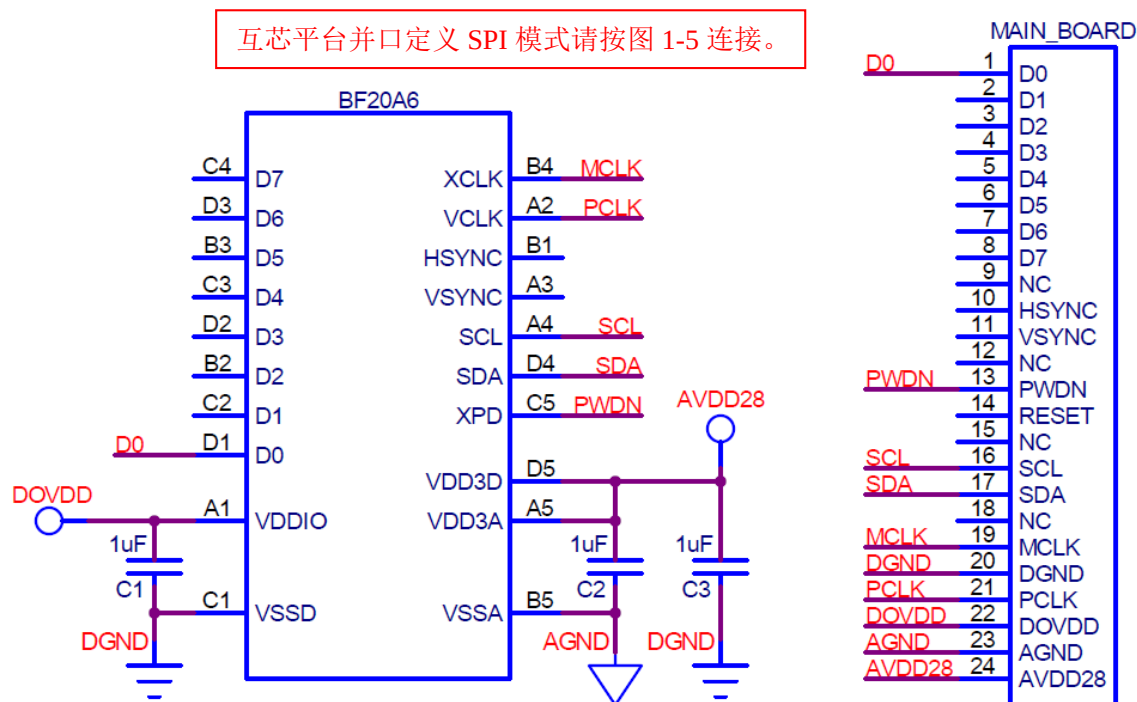


图 1-4 SPI 1-bit 模式外围电路图

1.5 互芯平台 SPI 模式（主板提供并口定义）

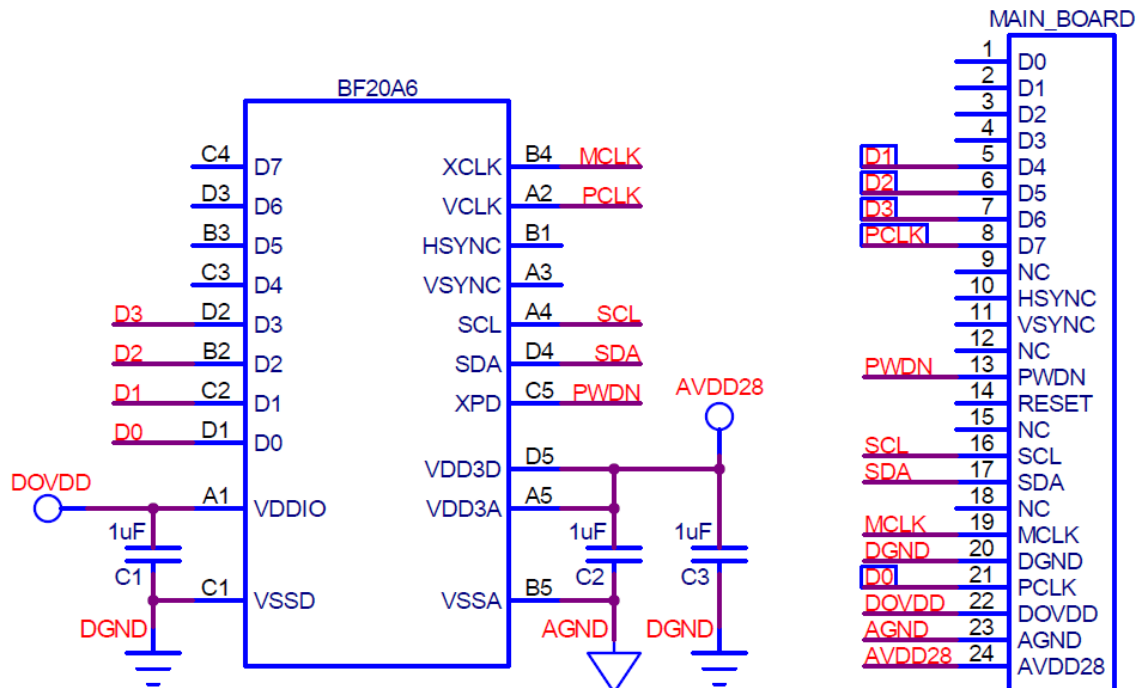


图 1-5 互芯平台并口定义 SPI 模式外围电路图

1.6 外围电路设计说明

- BF20A6 芯片分三路电源供电：
 - VDD3A, 2.7V~3.1V
 - VDD3D, 2.7V~3.1V
 - VDDIO, 1.7V~3.1V
- 主板 IO 电源 DOVDD 为 1.8V 时，芯片 VDDIO 必须连接到主板 IO 电源 DOVDD；
- 数字地与模拟地需要分开走线，当主板没有区分数字地和模拟地时，建议将地从靠近连接器位置分开，然后分别接到芯片数字地和模拟地；
- 电源线和地线至少需加宽到 0.2mm 以上，其它信号线宽需至少 0.1mm（受锡球间距限制部分除外），背面铺地铜，时钟线需两侧包地，FPC 较长或 SPI 模式应用时，在此基础上适当加宽线路；
- C1, C2, C3 省略会影响影像质量。C1 需靠近 A1(VDDIO)管脚，C2 需靠近 A5(VDD3A)，C3 需靠近 D5(VDD3D)管脚。

2 封装管脚

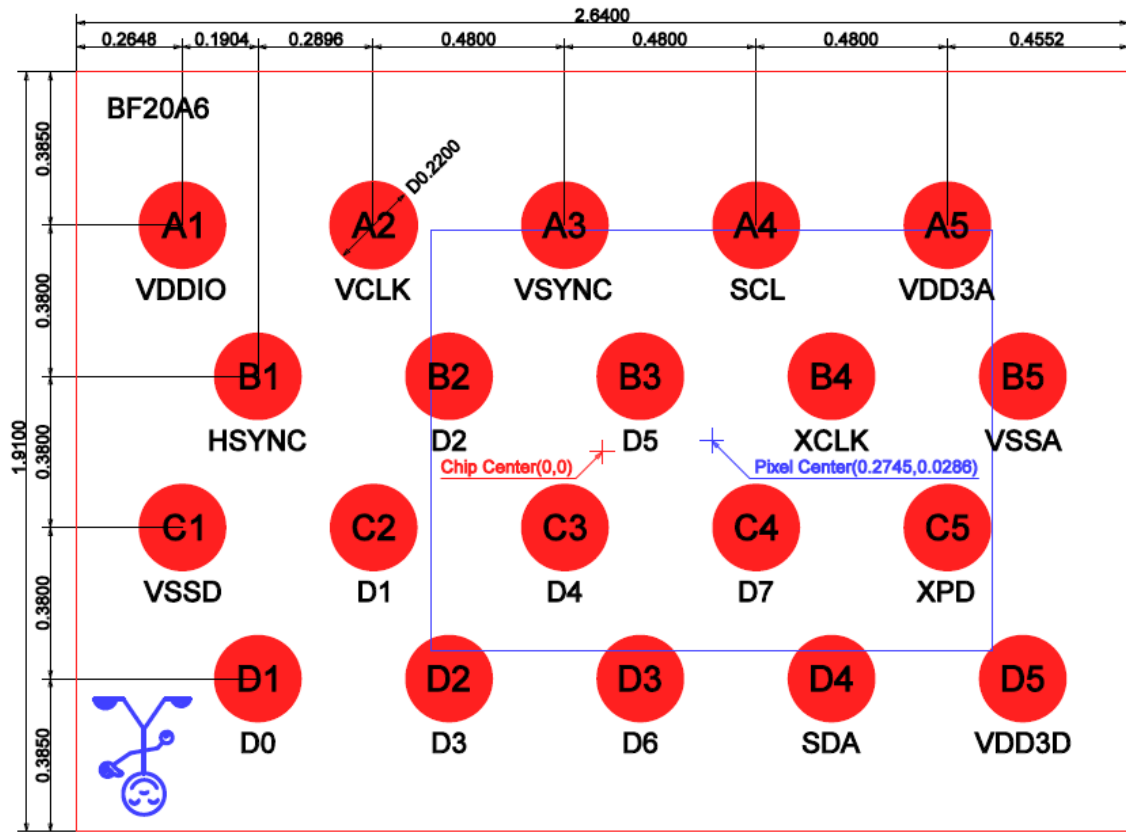


图 2-1 封装管脚透视图（单位：mm）

3 管脚说明

Pin	Name	Pin type	Function/Description
A1	VDDIO	Power	IO 电源（1.7-3.1V）
A2	VCLK	Output	视频采样时钟输出信号
A3	VSYNC	Output	VSYNC 输出信号
A4	SCL	Input	两线串行接口输入时钟信号
A5	VDD3A	Power	模拟电源（2.7V-3.1V）
B1	HSYNC	Output	HSYNC 输出信号
B2	D2	Output	YUV/RGB 图像数据输出端口 2
B3	D5	Output	YUV/RGB 图像数据输出端口 5
B4	XCLK	Input	XCLK 时钟输入信号
B5	VSSA	Power	模拟地
C1	VSSD	Power	数字地
C2	D1	Output	YUV/RGB 图像数据输出端口 1

C3	D4	Output	YUV/RGB 图像数据输出口 4
C4	D7	Output	YUV/RGB 图像数据输出口 7
C5	XPD	Input	芯片休眠模式控制 0: 正常工作模式 1: 休眠模式
D1	D0	Output	YUV/RGB 图像数据输出口 0
D2	D3	Output	YUV/RGB 图像数据输出口 3
D3	D6	Output	YUV/RGB 图像数据输出口 6
D4	SDA	I/O	两线串行接口数据 I/O 信号
D5	VDD3D	Power	数字电源 (2.7V-3.1V)

Note:

XPD : pull-down to VSSD by inner resistor.

4 封装规格

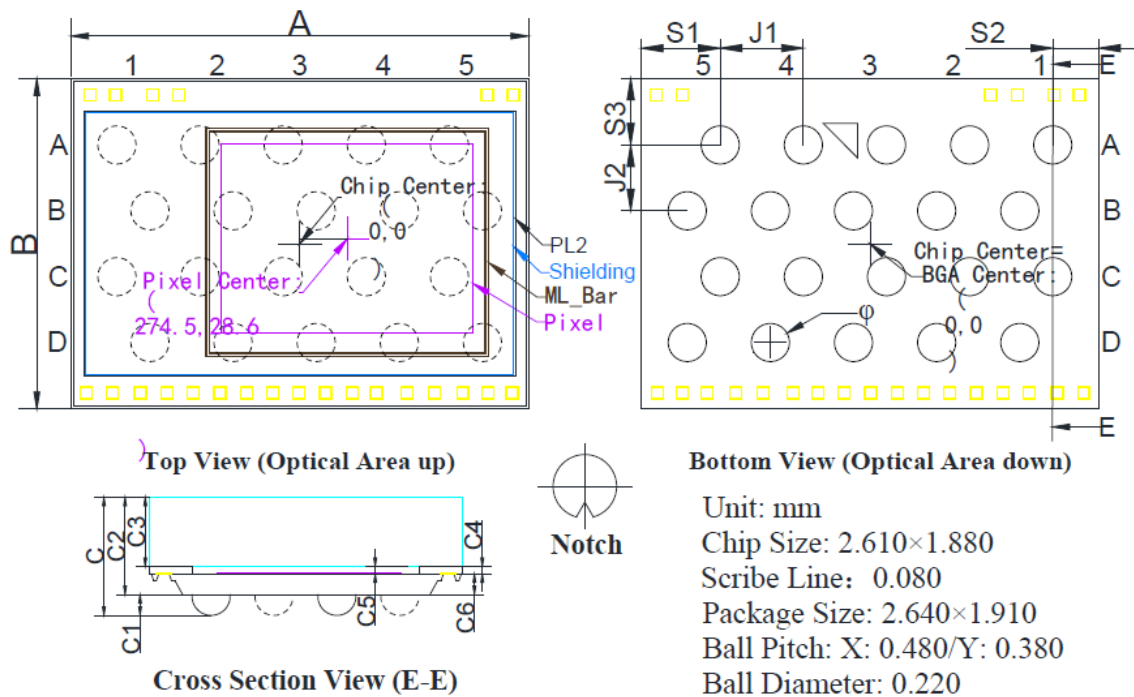


图 4-1 封装规格图

5 封装尺寸

描述	标识	典型值	最小值	最大值
		单位: mm		
封装体 X 方向尺寸	A	2.6400	2.6150	2.6650
封装体 Y 方向尺寸	B	1.9100	1.8850	1.9350
封装高度	C	0.6900	0.6350	0.7450
锡球高度	C1	0.1200	0.0900	0.1500
封装体厚度	C2	0.5700	0.5400	0.6000
玻璃厚度	C3	0.4000	0.3900	0.4100
CV 高度	C4	0.0300	0.0260	0.0340
CV+环氧树脂厚度	C5	0.0325	0.0275	0.0375
硅厚	C6	0.1000	0.0950	0.1050
锡球直径	D	0.220	0.190	0.250
管脚数量	N	20		
X 方向锡球间距	J1	0.4800	0.4700	0.4900
Y 方向管脚间距	J2	0.3800	0.3700	0.3900
X 方向管脚中心到边缘距离	S1	0.4552	0.4252	0.4852
X'方向管脚中心到边缘距离	S2	0.2648	0.2348	0.2948
Y'方向管脚中心到边缘距离	S3	0.3850	0.3550	0.4150

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